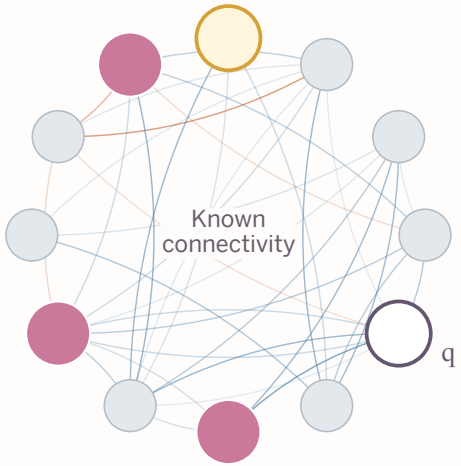
# Predictive risk and recording decisions

## in partially observed neural networks




**100 simulated units · M = 5**

Uncertain neuronal gains

- Recorded node
- Stimulated node
- Response readout (one of three)

### 1 Endpoint recording

Stimulation site and readouts

**Lower held-out error**

**98%** 37 of 40 networks improved



Both endpoints vs neither, mean MSE

`-98.13% [-99.19, -93.94]`

**Narrower feasible range**

**99%** 40 of 40 networks narrowed



Mean squared width, both vs neither

`-99.08% [-99.62, -96.10]`

Network not improved

### 2 Error factors

Endpoint vs ordinary recording
Exact three-factor split

**Input factor**

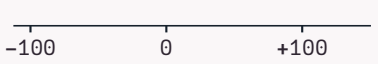
`-98.7%  [-99.5, -96.3]`

**Readout factor**

`-98.8%  [-99.6, -96.9]`

**Recurrent factor**

`+16.5%  [-34.6, +139.4]`




Mean-square change vs ordinary (%)

95% paired network-bootstrap CI

### 3 Fifth recording

Endpoints kept, one site added
80 new independent networks

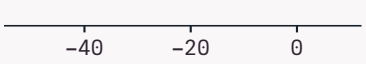
**63** of 80 networks improved combination estimates



Fifth site chosen for recurrent gain sums, compared with a random fifth site

**Combination estimation MSE**

`-31.1%  [-42.4, -19.8]`




Change vs random fifth site (%)

95% paired network-bootstrap CI

### Integrated stimulus response

$$q = \int \Delta r(t)\, dt$$

$$\Delta r(t) = r_{\text{stim}}(t) - r_{\text{control}}(t)$$

### Better risk estimates do not necessarily make better recording decisions.

We used a **common MAP estimator**, so identical recordings give identical predictions in **68% and 65%** of designs.
Recording the stimulation site and readouts lowered held-out response error by **98%** in independent networks.
A fifth site chosen for recurrent gain sums improved their estimates by **31%**, with no clear change in response error.

These simulations suggest that **endpoint coverage** is an informative baseline for evaluating recording criteria.